



Machine Learning Approach to Predicting Academic Dropout

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Abstract

Universities lose millions of dollars each year because of student dropout. Still, most universities do not have a system to catch students who are struggling academically. Advisors usually rely on signals, such as a low GPA and by the time they notice, the student is often already behind. Machine learning research increasingly shows that this predictive gap can be closed. Datasets already exist that contain socioeconomic and academic information needed for such prediction. The problem is that these datasets rarely label outcomes as dropout, enrolled or graduate.

This project uses machine learning on one dataset: the “Predict Students’ Dropout and Academic Success” dataset from the UCI repository. The dataset contains data for about 4,400 students, including socioeconomic information and grades from the first and second semesters. Each student is labelled as Dropout, Enrolled or Graduate. Therefore, this is a multi-class classification problem. We will compare methods. Methods include a naive-majority baseline, logistic regression, k-nearest neighbours, decision trees and ensemble methods. We will test all methods on a held-out sample. Use macro-average F1 as the evaluation metric. Additionally, we will perform subgroup error analysis to verify that students are treated consistently regardless of background factors such as qualifications or parental education. The goal is to show whether a systematic, evidence-based early-warning tool can be built to supplement the way advisors currently identify at-risk students.

Research Background

University dropout is still common in education. Many institutions say that first-year dropout rates can be between 20% and 30%. This causes financial and personal costs for both students and the university. This project’s task is to predict, using information on record early in a student’s time at the university, whether the student will drop out, stay enrolled or graduate.

Most universities still find at-risk students informally. The methods used include an instructor or advisor noticing missed classes. Applying a GPA cutoff after the student has already started struggling. This approach catches the problem late. Rarely uses demographic, socioeconomic or early-performance data that enrollment offices already have.

Automated dropout prediction is not new to educational data mining. Small teams seldom get to build, evaluate and interpret a full pipeline end-to-end on a public dataset within one course term. This project fills that gap by comparing several course-taught model families instead of reporting only one best-tuned result.

Methodology

Data: We will use the “Predict students’ Dropout and Academic Success” dataset from the UCI Machine Learning Repository (Realinho et al. 2021). The repository holds data for about 4,400 students. The data includes more than 30 features related to demographic information, admission grades, socioeconomic status and academic performance in the first and second semesters. Each record is labelled with one of three outcomes: dropout, enrolled or graduate. The dataset is free for researchers, educators and students and no additional data collection or human-subject approvals are needed.

Methods: We will build a majority-class baseline. We will also test a baseline (logistic regression), k-nearest neighbour, decision tree and an ensemble approach such as random forest or gradient boosting. All methods will use the preprocessed features and data split. We will split the data into

training, validation and test sets with stratification: about 70% for training, 15% for validation and 15% for testing. The validation set will guide the choice of the model and tuning of parameters. The test set will be used once for the final evaluation.

Significance

The main contribution is a benchmark. We want to see whether enrollment and first-semester data can actually predict dropout risk. We compare a baseline with several methods from this course, using one fixed evaluation protocol instead of only reporting the best run. Advising and retention offices can use this evidence to decide whether a systematic early-warning tool is worth building. If such a tool is adopted, students benefit from more consistent outreach. We also gain practice experience.

We want to compare groups of models using a clearly documented evaluation protocol rather than doing separate experiments each time.

Currently, spotting dropout is mostly reactive. An advisor sees a problem informally; practices differ across departments. When a student is flagged, they are often already having difficulty. This project offers an approach that uses data the registrar's office already gathers when students enroll.

We will declare the project a success if at least one trained model outperforms the simple baseline by a clear margin on macro-F1, if the entire pipeline can be rebuilt from the repository using the same data splits and random seeds, and if the subgroup error analysis shows no large unexplained gaps across attributes such as prior qualification or parental education. A positive outcome could be expanded to enrollment groups, additional features or a more thorough fairness audit. Even a negative or mixed outcome is valuable; it shows an advising office what methods do not work and can guide projects or lab work on this dataset.

SMART Timeline

Milestone 1 – Proposal & Team Contract (Sept 18, 2026)

Finalize the project proposal, submit the team contract, and confirm role assignments.

Milestone 2 – Dataset & Problem Approval (Oct 2, 2026)

Confirm dataset license/access, complete initial EDA, document class balance and missingness, and fix the train/val/test split and primary metric.

Milestone 3 – Baseline Reproduction (Oct 19, 2026)

Working data pipeline, naive benchmark, and course-level baseline implemented and documented with a baseline results table. Complete Start/Stop/Continue plan.

Milestone 4 – Progress Presentation (Oct 30, 2026)

Preliminary results from at least two additional models (k-NN, decision tree, and/or ensemble) and revised timeline based on progress to date.

Milestone 5 – Final Experiments & Analysis (Nov 20, 2026)

Complete fair comparison across all models, tuning procedure, uncertainty/sensitivity analysis, and draft results section.

Milestone 6 – Final Report, Repository & Demo (Dec 4, 2026)

Final 6-8 page report, tagged reproducible repository release, and final presentation/demo.

Milestone 7 – Individual Reflection & Oral Defence (Dec 7, 2026)

Submit individual reflections and oral defence.

References

Realinho, V., Vieira Martins, M., Machado, J., & Baptista, L. (2021). Predict Students' Dropout and Academic Success [Dataset]. UCI Machine Learning Repository. <https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success>